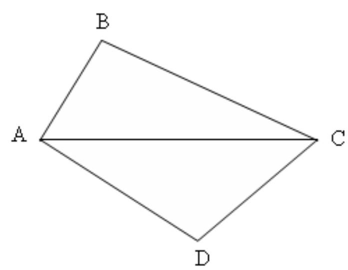
JMO – 2007

Max Mark – 100 Time – 3 Hrs

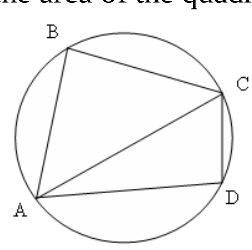
Instructions:

- Calculators (in any form) and protractors are not allowed.
- Ruler and compass allowed.
- Answer all the questions.

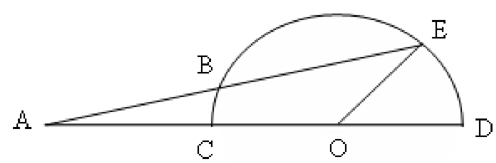
1. A merchant paid Rs 30 for an article. He wishes to place price tag on it so that he can offer a 10% discount on the price marked on the tag and still make a profit of 20 % on the cost. What price should he mark on the tag?
2. Find out the average of 1, 2, 3, ….., 999, 1000.
3. Manas have ten rupee note, twenty rupee note and fifty rupee note in his purse. He has atleast one ten rupee note and at least three twenty rupee notes. The total value of the notes is Rs 560. What is the smallest possible number of notes Manas can have in his purse?
4. Mohan has two brothers, Ajay and Dinesh. The product of the ages of all three children is 144. Ajay is 11 year older than Dinesh. How old is Mohan? Explain fully how you arrived at your answer?
5. What is the middle digit of the product $968880726456484032 \times 875$
6. Find area of the convex quadrilateral ABCD, given AB = 15, BC = 20, CD = 7, AD = 24 and AC = 25.



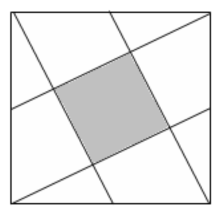
7. Find all the multiples of 10, between 1 and 2000, which are equal to the sum of two positive numbers, one of which is a perfect square and other is a perfect cube.
8. There was a cricket match between India and Pakistan at Cuttack. Tickets for each boy student were Rs 10 and for each firl student were Rs. 6. A particular school in Cuttack wanted to pay for the tickets for all his students who wanted to see the cricket match. All girls of the school went to see the match but 40% of the boys of the school declined to see the match. There were a total of 2240 boys and girls in the school. How much did the school pay for the tickets?
9. Mr. Chotray has five children all at least one year old. The oldest child is aged nine, and no two are of the same age in years. The middle child's age is one quarter of the total of all the children's ages. What are the ages of the children?
10. The points A, B, C and D are placed consecutively on a circle with diameter AC so that AB = 7 cm, BC = 24 cm and CD = 15 cm. What is the area of the quadrilateral ABCD?



11. In a division, the dividend is 4758, the quotient is 25 and the remainder is more than 25 but less than 50. Find out the divisor.
12. A boy was asked to find out LCM of 27, 45 and a third number. While writing these numbers he wrote 72 in place of 27 but wrote 45 and the third number correctly. However he got the same LCM as those of original numbers. What is the smallest possible value of the third number and what is the LCM of these numbers?
13. If we add 5 to the ten's digit and subtract 3 from the unit digit of a two digit number, then resulting number is twice the original number. Find out the original two digit number?
14. You have a frying pan which will only take two slices of bread at a time and you with to fry three slices, each on both sides. Since each slice takes 20 seconds for each side, you can certainly fry them all in 80 seconds by doing two pieces together and then the third. But can you fry them all in less than 80 seconds? Explain your answer.
15. The pages of a book are numbered consecutively: 1, 2, 3, 4, …., 11, 12, …. And so on. No pages are missing. If in page numbers the digit 3 occurs 99 times what is the number of the last page?
16. A certain 7 digit number 1234***, where each * represents a missing digit, is a perfect square. Find its square root. Explain with reasoning.
17. There are 24 four digit numbers using the digits 1, 3, 5, 7 once in each. What is their sum?
18. In the figure CD is a diameter of the semicircle with centre at O, AB = OD and $m\angle DOE = 45^0$, then compute $m\angle BOE$.



19. In the diagram at right segments join the vertices of a unit square to the midpoints of its sides. Find the area of the shaded quadrilateral?



20. Find which is greater 65^{1662} $or\, 33^{1995}$,without using Calculator.

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